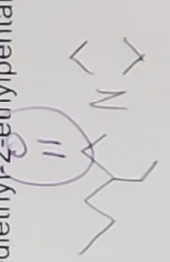
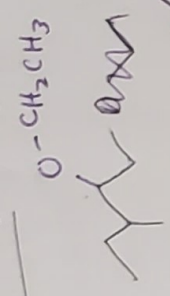
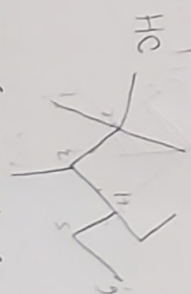
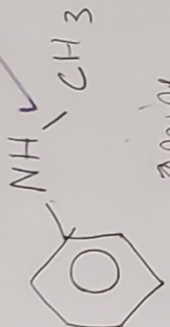
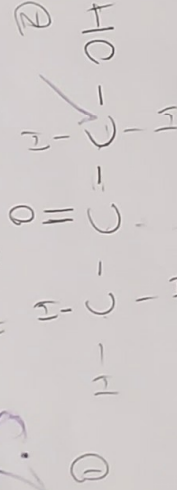
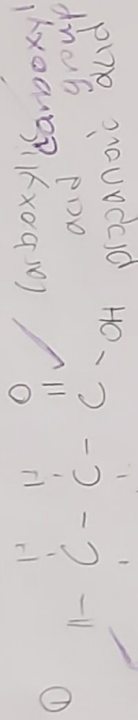


Communication/Application (15 Marks Total)

1. Complete the table below by naming the structural diagram for each of the organic compounds or drawing a structural diagram or carbon skeleton for each organic compound. (4 marks)

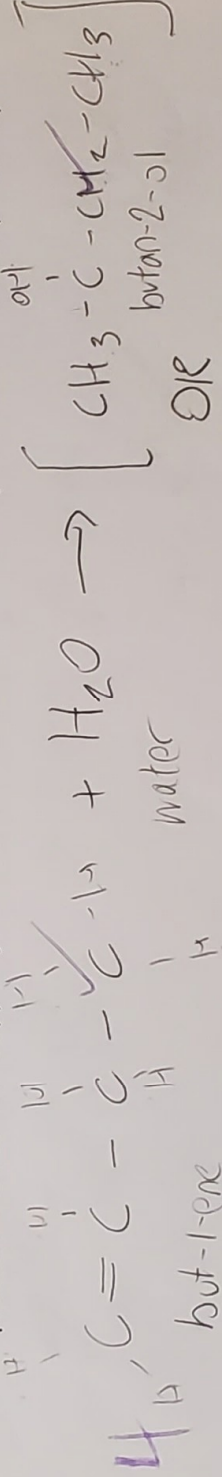
(a) N,N-diethyl-2-ethylpentanamide	(b) 3-ethoxy-4-methylhexane
 amide	 ether
(c) 4-ethyl-2,3-dimethylhexan-2-ol	(d) N-methylaniline
 alcohol	 amine

2. Draw 2 isomers for the following formula, $C_3H_6O_2$. Then, (1) give the IUPAC name of each, (2) identify the functional group to which each belongs, and (3) determine which molecule is more soluble in a non-polar solvent. (3 marks)



③ The ketone is more soluble because it is less polar than the acid, and since the its less polar it dissolves better in non-polar solvents.

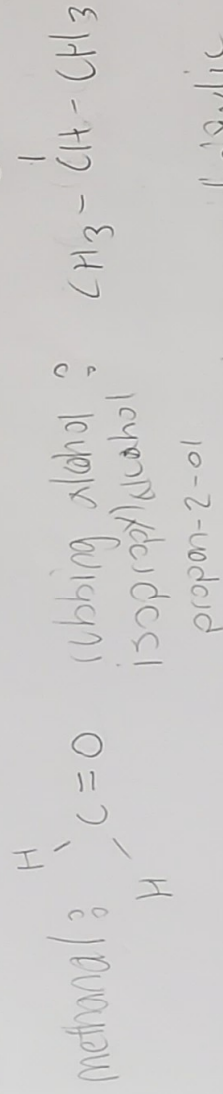
3. The reaction of but-1-ene with water produces an alcohol; however, two different products can be produced in this reaction. Complete each reaction by drawing and naming the products of each. Which product is favoured? Support your answer with reference to concepts learned in class. (4 marks)



Butan-2-ol is the preferred product due to Markovnikov's rule which states that the carbon in a bond is more likely to get the hydrogen if it has more hydrogens than the other carbon in the bond. Since the end has more H's, it receives the H⁺ from H₂O, while the other receives the OH⁻.

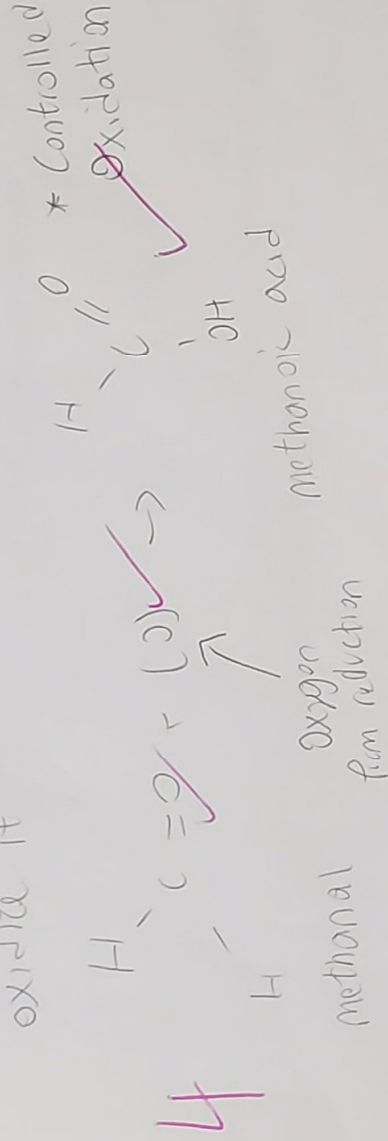
the hydrogen if it has more hydrogens than the other carbon in the bond. Since the end has more H's, it receives the H⁺ from H₂O, while the other receives the OH⁻.

You want to make a lovely perfume for your mother, and since you are a chemistry student you decide to make an ester. If you buy formaldehyde (methanal) and rubbing alcohol (isopropyl alcohol) from a science shop, what would your ester be called? Write a series of reactions showing how you would do this. (4 marks)

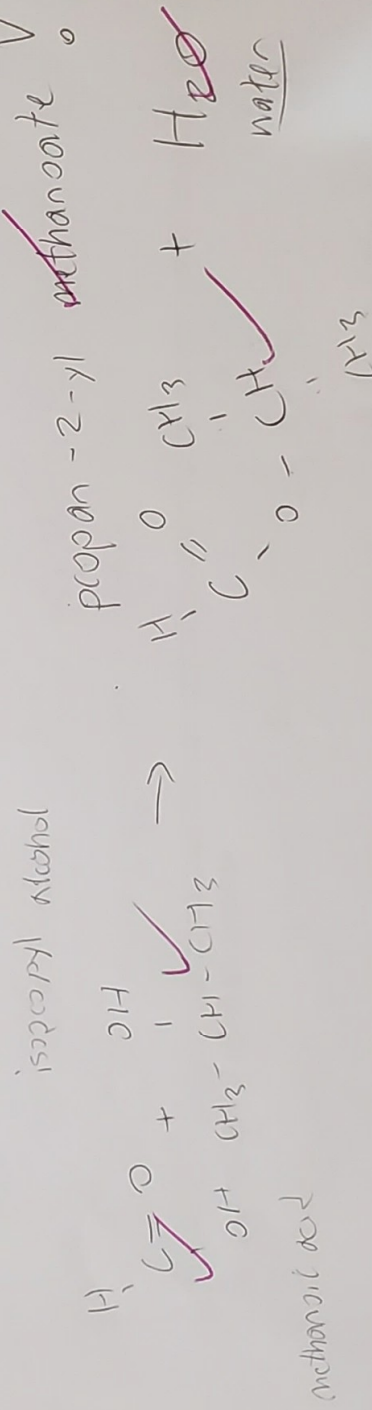


To make an ester we need an alcohol and a carboxylic acid.
To make a carboxylic acid we can take the aldehyde and

oxidize it

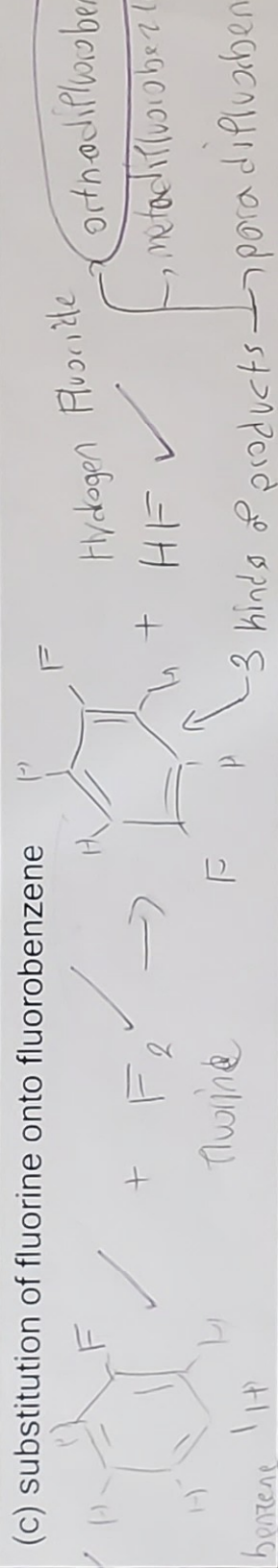
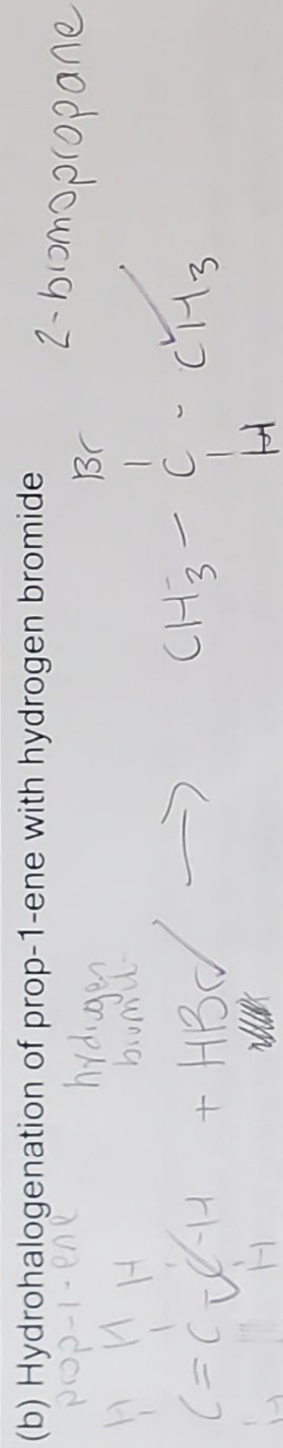
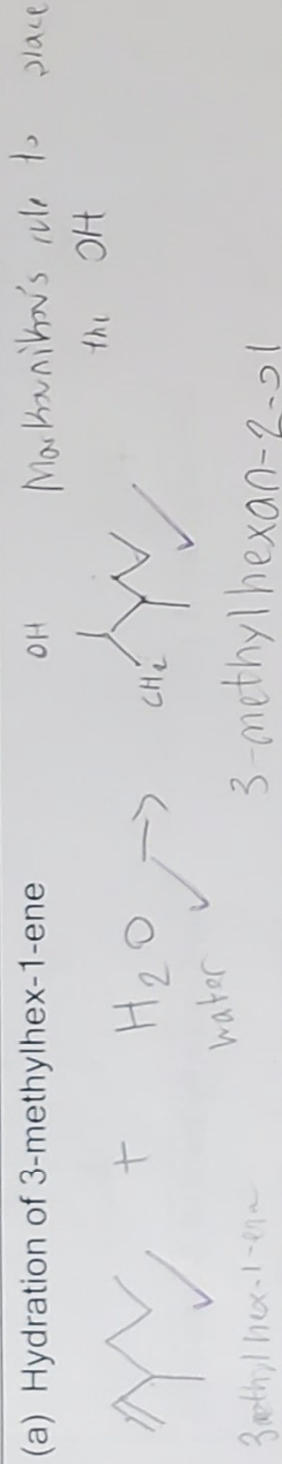


We then combine methanoic acid with isopropyl alcohol to make...

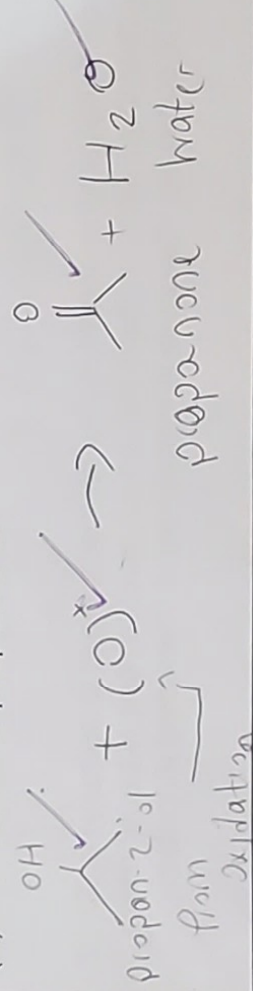


So the ester is propan-2-yl methanoate

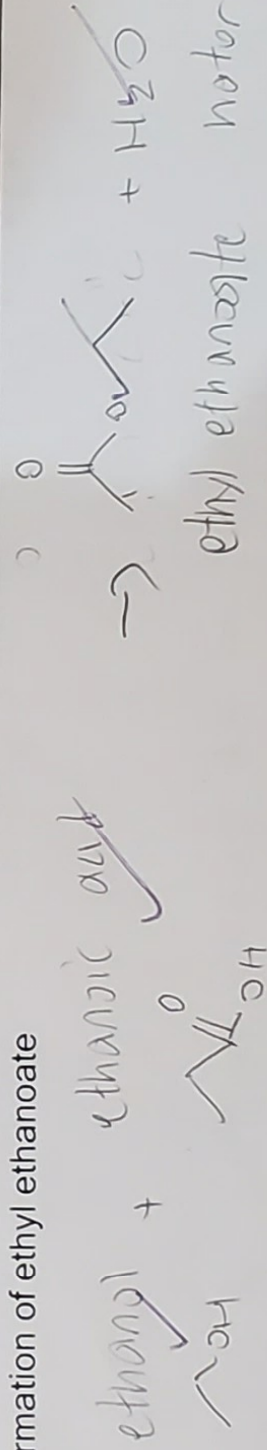
1. Write an equation, using structural formula or carbon skeleton, for each of the following reactions and name the reactants and products. (18 MARKS TOTAL - 3 MARKS FOR EACH REACTION)



- (d) Oxidation of propan-2-ol



- (e) Formation of ethyl ethanoate



- (f) Formation of N-methyl butanamide

